CyberShield AI - Mini Cybersecurity Toolkit

Abstract

This project presents CyberShield AI, a lightweight cybersecurity toolkit designed to detect phishing attempts, analyze suspicious URLs, and identify basic anomalous login behavior patterns. The system demonstrates practical security awareness using simple rule-based and pattern-based detection techniques.

Problem Statement

With the increasing number of cyber threats such as phishing emails, malicious links, and brute-force login attacks, there is a need for simple yet effective tools that help identify suspicious activity at an early stage.

Objectives

- Detect phishing emails using keyword and pattern analysis
- Identify suspicious URLs based on structural indicators
- Detect repeated login failures as a sign of brute-force attempts

System Design

The system is designed as a modular toolkit with three components: Email Analyzer, URL Checker, and Log Analyzer. Each module processes input data and returns a risk classification.

Methodology

The project uses rule-based logic and pattern recognition techniques. Inputs are analyzed for known phishing indicators such as urgency words, suspicious domains, excessive numeric patterns, and repeated access failures.

Tools Used

Python, Google Colab / Jupyter Notebook, basic string processing techniques, optional Streamlit for UI

Expected Output

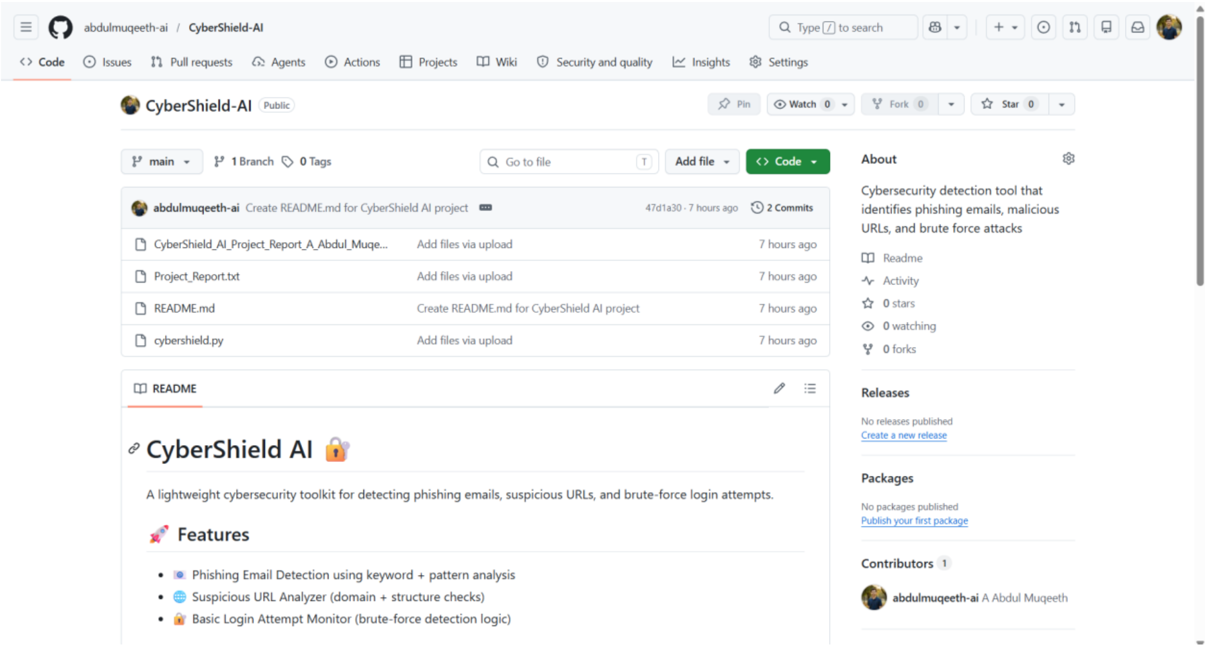
The system classifies inputs into Safe, Suspicious, or High Risk categories and provides explanations for each classification.

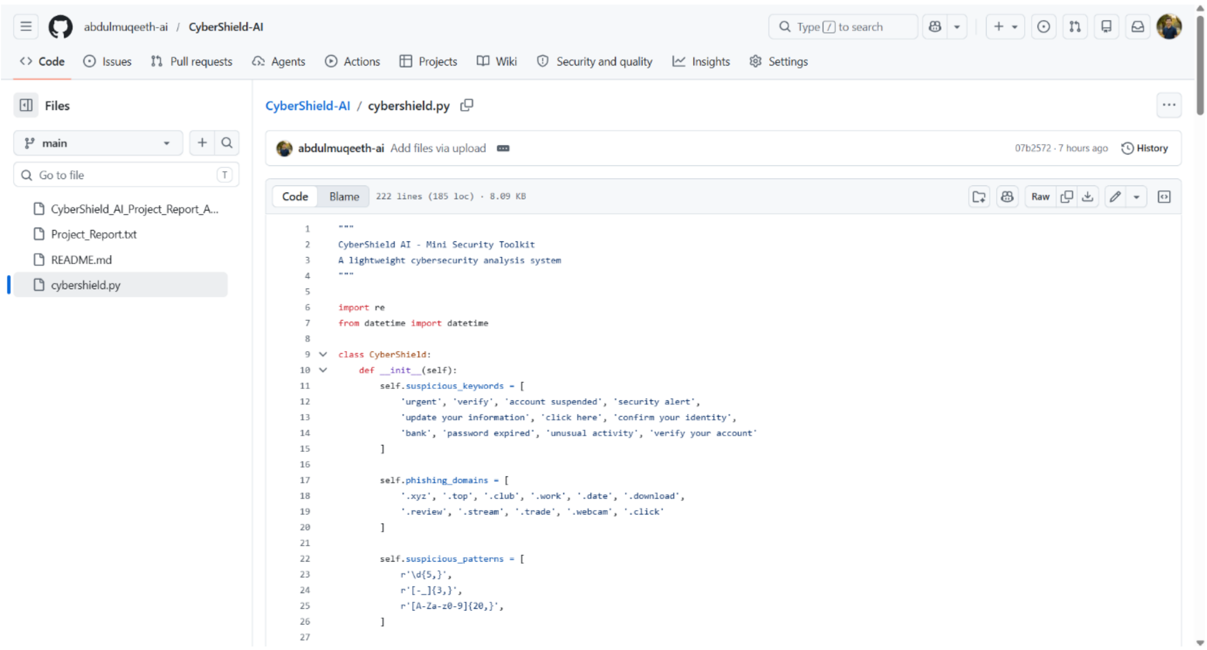
Conclusion

CyberShield AI demonstrates how basic cybersecurity concepts can be implemented in a simple system to improve awareness and early threat detection. It is a foundational step toward advanced AI-based security systems.

GitHub

Repository link: https://github.com/abdulmuqeeth-ai/CyberShield-AI


abdulmuqeeth-ai / CyberShield-AI
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
CyberShield-AI Public
Pin Watch 0 Fork 0 Star 0
main 1 Branch 0 Tags
Go to file
Add file
Code
abdulmuqeeth-ai Create README.md for CyberShield AI project 47d1a30 · 7 hours ago 2 Commits
CyberShield_AI_Project_Report_A_Abdul_Muqe... Add files via upload 7 hours ago
Project_Report.txt Add files via upload 7 hours ago
README.md Create README.md for CyberShield AI project 7 hours ago
cybershield.py Add files via upload 7 hours ago
README
CyberShield AI
A lightweight cybersecurity toolkit for detecting phishing emails, suspicious URLs, and brute-force login attempts.
Features
Phishing Email Detection using keyword + pattern analysis
Suspicious URL Analyzer (domain + structure checks)
Basic Login Attempt Monitor (brute-force detection logic)
About
Cybersecurity detection tool that identifies phishing emails, malicious URLs, and brute force attacks
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
abdulmuqeeth-ai A Abdul Muqeeth


abdulmuqeeth-ai / CyberShield-AI
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
Files
main
Go to file
CyberShield_AI_Project_Report_A...
Project_Report.txt
README.md
cybershield.py
CyberShield-AI / cybershield.py
abdulmuqeeth-ai Add files via upload 07b2572 · 7 hours ago History
Code Blame 222 lines (185 loc) · 8.09 KB
Raw
"""
CyberShield AI - Mini Security Toolkit
A lightweight cybersecurity analysis system
"""
import re
from datetime import datetime
class CyberShield:
def __init__(self):
self.suspicious_keywords = [
'urgent', 'verify', 'account suspended', 'security alert',
'update your information', 'click here', 'confirm your identity',
'bank', 'password expired', 'unusual activity', 'verify your account'
]
self.phishing_domains = [
'.xyz', '.top', '.club', '.work', '.date', '.download',
'.review', '.stream', '.trade', '.webcam', '.click'
]
self.suspicious_patterns = [
r'\d{5,}',
r'[-_]{3,}',
r'[A-Za-z0-9]{20,}',
]